Two-row $d = 4$ law, classes $b \equiv 2, 3 \pmod 4$: the unique 2-adic root and a single-candidate certificate

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Status. ($\star$) *not closed uniformly.* The bare-factor-m "obstruction" is shown to make Q_b have a *unique* root in $\mathbb{Z}_2$, so Q_b has *at most one* rational root. This yields the cleanest known certificate (one candidate, one prime, one evaluation per b), with which ($\star$) is verified for all $b \equiv 2, 3 \pmod 4$ in $[6, 171]$. The uniform gap is reduced to a single 2-adic statement.

1 The object and the gap

For $\lambda = (2m - b, b)$, $s = 1 + i$, $A = 1 + su + u^2$, set $g_b(m) = [u^b]A^m$, $G_b(m) = [u^b]((1 - u)A^m) = g_b - g_{b-1}$, and $I_b(m) = \operatorname{Im} G_b(m) \in \mathbb{Z}[m]$ (degree b). The two-row $d = 4$ fiber law $G_{(2m-b,b)}(i) = 0 \iff (2, 2)$ reduces, after dividing out the forced integer roots $m = 0, 1, \ldots, R$ ($R = \lfloor (b-1)/2 \rfloor$), to the monic-over-$\mathbb{Q}$ polynomial $Q_b(m)$ of degree $d = \lfloor b/2 \rfloor$, and to

$$(\star) \qquad \text{for } b \equiv 2, 3 \pmod 4,\ b \geq 3 : \quad Q_b(m) \text{ has no integer root } m \geq b.$$

Classes $b \equiv 0, 1 \pmod 4$ are already proved unconditionally. Closing ($\star$) finishes the law.

2 The engine

Writing $A = (su) + (1 + u^2)$ and expanding gives $g_b(m) = \sum_j \binom{m}{j} s^j \binom{m-j}{(b-j)/2}$ (only $j \equiv b \bmod 2$). Subtracting g_{b-1}:

$$\boxed{I_b(m) = \sum_{j=1}^{b} (-1)^{b-j} \operatorname{Im}(s^j) \binom{m}{j} \binom{m-j}{\lfloor (b-j)/2 \rfloor}}, \quad \operatorname{Im}(s^j) = \begin{cases} 0, & 4 \mid j, \\ \{1, 2, 2\}[j \bmod 4]\,(-4)^{\lfloor j/4 \rfloor}, & \text{else.} \end{cases}$$

Verified equal to the direct extraction $\operatorname{Im}[u^b]((1 - u)(1 + su + u^2)^m)$.

3 The mod-2 collapse and the unique 2-adic root

Lemma 1 (mod-2 collapse)**.** *$I_b(m) \equiv m\binom{m-1}{R} \pmod 2$, $R = \lfloor (b-1)/2 \rfloor$.*

Proof. For $j \geq 4$, $(-4)^{\lfloor j/4 \rfloor} \equiv 0 \pmod 2$; for $j \in \{1, 2, 3\}$, $\operatorname{Im}(s^j) = 1, 2, 2 \equiv 1, 0, 0 \pmod 2$. So mod 2 only $j = 1$ survives: $I_b \equiv (-1)^{b-1}\binom{m}{1}\binom{m-1}{R} \equiv m\binom{m-1}{R}$. $\square$

The bare factor m blocked every prior single-prime attempt (root $m \equiv 0$); but it makes the 2-adic root *unique.*

Lemma 2 (mod-2 factorisation; prior cycle via Mahler coefficients, re-verified $6 \le b \le 80$)**.** *For the monic, 2-integral Q_b, $Q_b \equiv m\,(m^2+m+1)^{(d-1)/2} \pmod 2$, with m^2+m+1 irreducible over $\mathbb{F}_2$ and $(d-1)/2 = \lfloor (b-1)/4 \rfloor \in \mathbb{Z}$.*

Theorem 1 (unique 2-adic root)**.** *For every $b \equiv 2,3 \pmod 4$, Q_b has exactly one root $\rho_b \in \mathbb{Z}_2$, and ρ_b is even.*

Proof. By the Lemma, over $\mathbb{F}_2$ the only root of Q_b is $m = 0$ (m^2+m+1 has no $\mathbb{F}_2$-root, its values at $0, 1$ being 1), and it is *simple*: $(Q_b \bmod 2)'(0) = (m^2+m+1)^{(d-1)/2}|_{m=0} = 1 \neq 0$. Hensel's lemma lifts it to a unique $\rho_b \in \mathbb{Z}_2$ with $\rho_b \equiv 0 \pmod 2$. Conversely any $\mathbb{Z}_2$-root reduces mod 2 to an $\mathbb{F}_2$-root, hence to $m = 0$, hence (Hensel uniqueness) equals ρ_b. So ρ_b is the unique $\mathbb{Z}_2$-root and is even. □

(The 2-adic Newton polygon [slope -1, run 1] $\oplus$ [slope 0, run $d-1$], computed for $b \le 31$, sharpens $v_2(\rho_b) = 1$ and places the unit roots in the unramified quadratic extension; only $v_2(\rho_b) \ge 1$ is needed below.)

Corollary 1. *Q_b has at most one rational root, namely ρ_b, which is even.*

Proof. A rational root is an integer (monic), hence lies in $\mathbb{Z} \subset \mathbb{Z}_2$ and is a root there; by the theorem it equals ρ_b. □

This inverts the obstruction: the bare m collapses the candidate set from $\sim 0.1\,b^2$ integers to a *single* number ρ_b, computable from the single prime 2.

4 The single-candidate certificate

Theorem 2 (certificate)**.** *Fix $b \equiv 2,3 \pmod 4$. Let $U_b = 1 + \max_i |a_i|$ be the Cauchy bound on the roots of monic $Q_b = \sum a_i m^i$. Choose K with $2^K > U_b$ and lift the 2-adic root to $r := \rho_b \bmod 2^K \in [0, 2^K)$. Then the only possible integer root $m_0 \in [b, U_b]$ is $m_0 = r$; if $r \notin [b, U_b]$ or $Q_b(r) \neq 0$, then Q_b has no integer root $\ge b$, i.e. $(\star)$ holds for b.*

Proof. An integer root $m_0 \ge b$ has $0 < m_0 \le U_b < 2^K$ and $m_0 = \rho_b$ in $\mathbb{Z}_2$, so $m_0 \equiv \rho_b \pmod{2^K}$; its unique representative in $[0, 2^K)$ is r, whence $m_0 = r$. One exact evaluation $Q_b(r)$ settles it. □

Theorem 3 (frontier)**.** *$(\star)$ holds for all $b \equiv 2,3 \pmod 4$ with $6 \le b \le 171$; hence the two-row $d = 4$ law holds for these b. Cross-checked against exact rational-root enumeration for $b \le 40$.*

Examples: $b = 6$, $U_6 = 54$, sole candidate $r = 18$, $Q_6(18) = 1920 \neq 0$; $b = 7$, $U_7 = 791$, sole candidate $r = 610$, $Q_7(610) \neq 0$.

5 The residual gap

> **Open ($(\star)$, uniform).** For all $b \equiv 2,3 \pmod 4$: the unique 2-adic root $\rho_b \in \mathbb{Z}_2$ is not a rational integer in $[b, \infty)$. Equivalently, Q_b has no linear factor over $\mathbb{Q}$, i.e. irreducibility (or just no-linear-factor) of the Gegenbauer-in-parameter family $g_b(m) = C_b^{(-m)}(-(1+i)/2)$.

By the corollary this is a statement about *one* 2-adic number per b. For irreducible Q_b (empirically all, $b \leq 171$) it is automatic, and Jordan's derangement theorem + Chebotarev furnish a positive density of rootless primes (finite certificates). Rigorously dead and not to be retried: single-prime Newton polygon (integral slopes at 2; no full non-integral run for $p \leq 7$), uniform witness prime, finite covering set, analytic/log-concavity domination (in-range real roots), and the q-Lucas / Gaussian-binomial shortcut.